CLI Usage

1. Using CLI for Prediction

CLI (Command-Line Interface) is a way for users to interact with computers or software programs, where users run programs or execute tasks by typing text commands, rather than clicking icons or buttons through a graphical user interface (GUI).

2. CLI Usage Syntax

```
yolo TASK MODE ARGS

Where   TASK (optional) is one of [detect, segment, classify, pose, obb]
        MODE (required) is one of [train, val, predict, export, track,
benchmark]
        ARGS (optional) are any number of custom 'arg=value' pairs like
'imgsz=320' that override defaults.
```

3. Image Prediction

Use yolo26n.pt to predict the built-in images in the ultralytics26 project: If the system cannot find the corresponding model file in the directory where the command is run, it will automatically download (if download fails, you can copy the model in yourself)

Enter the project folder:

```
cd /home/sunrise/ultralytics26/
```

Use yolo26n.pt to detect images in the target folder and output results:

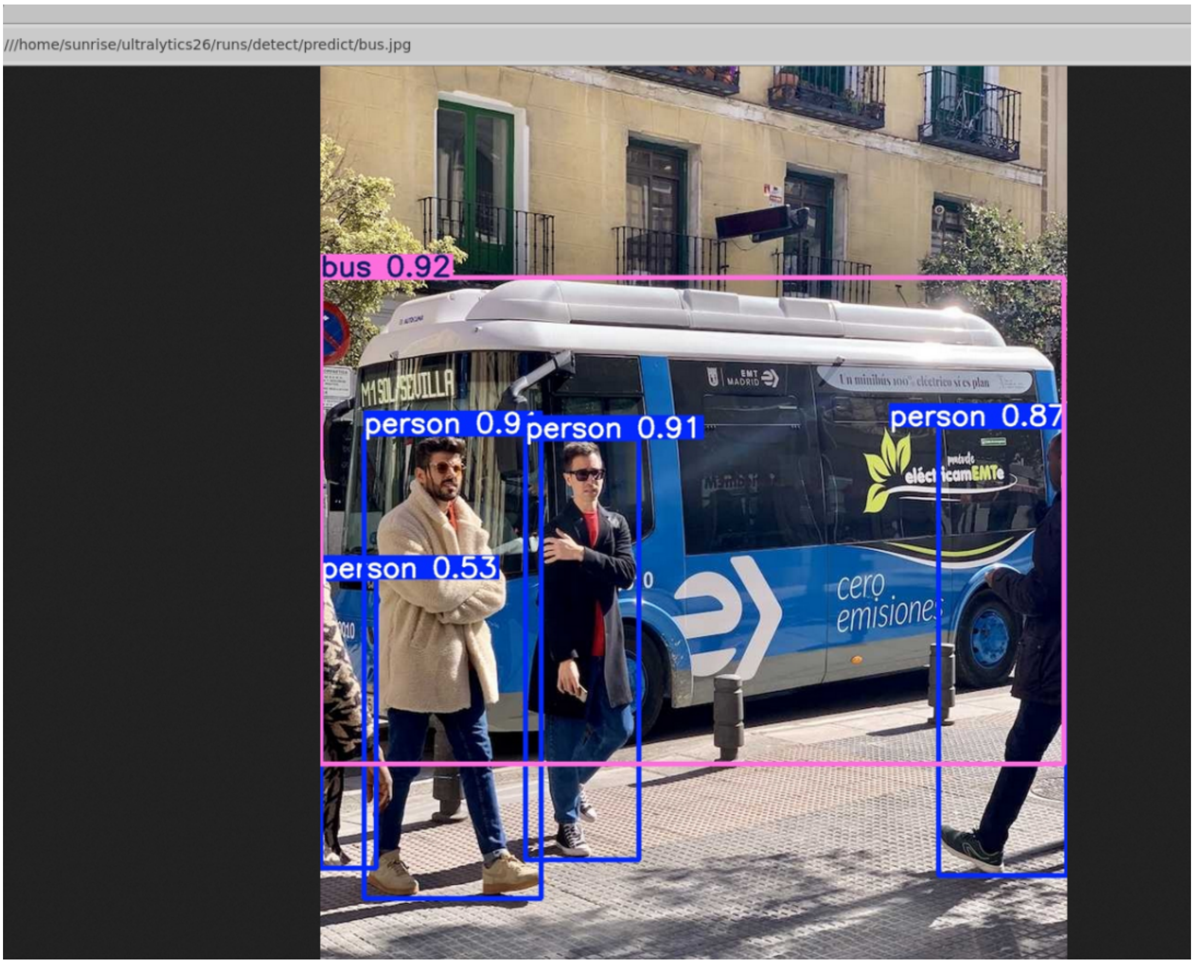
```
yolo predict model=yolo26n.pt source='/home/sunrise/ultralytics26/assets'
```

```
image 1/5 /home/sunrise/ultralytics26/assets/bus.jpg: 640x480 4 persons, 1 bus,
1300.2ms
image 2/5 /home/sunrise/ultralytics26/assets/car.jpg: 640x640 1 bus, 1646.9ms
image 3/5 /home/sunrise/ultralytics26/assets/dog.jpg: 480x640 2 dogs, 1233.9ms
image 4/5 /home/sunrise/ultralytics26/assets/people.jpg: 448x640 2 persons, 1 be
nch, 1208.4ms
image 5/5 /home/sunrise/ultralytics26/assets/zidane.jpg: 384x640 2 persons, 1 ti
e, 1108.8ms
Speed: 28.2ms preprocess, 1299.6ms inference, 1.2ms postprocess per image at sha
pe (1, 3, 384, 640)
Results saved to /home/sunrise/runs/detect/predict
```

Effect Preview

yolo recognition output video location: /home/sunrise/ultralytics26/runs/detect/predict

Each time it runs, the predict under this folder automatically increments by 1



4. Video Prediction

Use yolo26n.pt to predict videos in the ultralytics project (not videos that come with ultralytics): If the system cannot find the corresponding model file in the directory where the command is run, it will automatically download (if download fails, you can copy the model in yourself)

Enter the project folder:

```
cd /home/sunrise/ultralytics26
```

Use yolo26n.pt to detect videos in the target folder and output results:

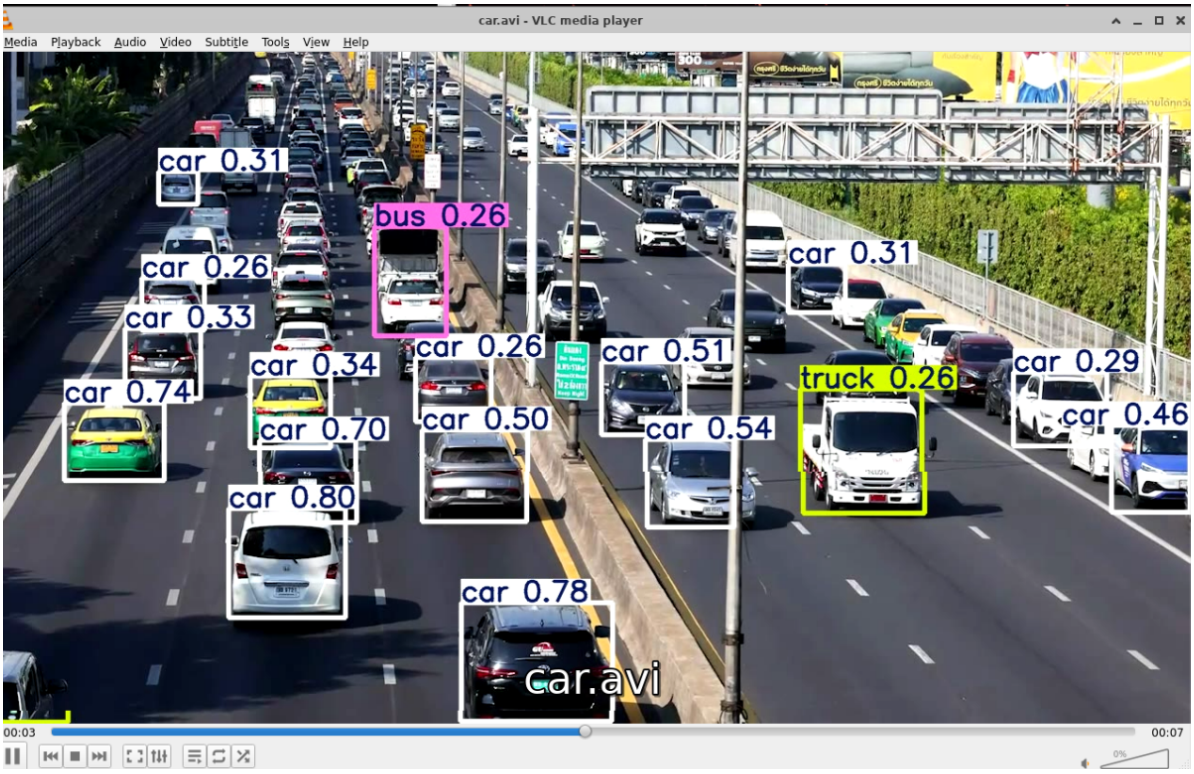
```
yolo predict model=yolo26n.pt source='/home/sunrise/ultralytics26/videos'
```

```
video 1/5 (frame 57/180) /home/sunrise/ultralytics26/videos/car.mp4: 384x640 16 cars, 2 trucks, 1027.7ms
video 1/5 (frame 58/180) /home/sunrise/ultralytics26/videos/car.mp4: 384x640 16 cars, 2 trucks, 998.8ms
video 1/5 (frame 59/180) /home/sunrise/ultralytics26/videos/car.mp4: 384x640 15 cars, 2 trucks, 1013.6ms
video 1/5 (frame 60/180) /home/sunrise/ultralytics26/videos/car.mp4: 384x640 1 person, 15 cars, 2 trucks, 999.1ms
video 1/5 (frame 61/180) /home/sunrise/ultralytics26/videos/car.mp4: 384x640 14 cars, 2 trucks, 1010.4ms
video 1/5 (frame 62/180) /home/sunrise/ultralytics26/videos/car.mp4: 384x640 1 person, 15 cars, 2 trucks, 1003.3ms
video 1/5 (frame 63/180) /home/sunrise/ultralytics26/videos/car.mp4: 384x640 15 cars, 2 trucks, 1020.6ms
video 1/5 (frame 64/180) /home/sunrise/ultralytics26/videos/car.mp4: 384x640 15 cars, 2 trucks, 1028.4ms
video 1/5 (frame 65/180) /home/sunrise/ultralytics26/videos/car.mp4: 384x640 16 cars, 1 bus, 1 truck, 1050.2ms
video 1/5 (frame 66/180) /home/sunrise/ultralytics26/videos/car.mp4: 384x640 15 cars, 1 truck, 1085.0ms
```

Effect Preview

yolo recognition output video location: /home/sunrise/ultralytics26/runs/detect/predict2

Each time it runs, the predict under this folder automatically increments by 1



5. Real-time Prediction

Use yolo26n.pt to predict USB camera footage: If the system cannot find the corresponding model file in the directory where the command is run, it will automatically download (if download fails, you can copy the model in yourself)

Enter the project folder:

```
cd /home/sunrise/ultralytics26
```

Use yolo26n.pt to detect camera footage and output results: Currently only USB cameras can directly use CLI to predict real-time footage, no relevant information found for CSI cameras to be directly used as input source

```
yolo predict model=yolo26n.pt source=0 save=False show # Object detection
# Instance segmentation: yolo predict model=yolo26n-seg.pt source=0 save=False
show
# Image classification: yolo predict model=yolo26n-cls.pt source=0 save=False
show
# Pose estimation: yolo predict model=yolo26n-pose.pt source=0 save=False show
# Oriented object detection: yolo predict model=yolo26n-obb.pt source=0
save=False show
```

Click on the terminal and press "Ctrl + C" shortcut to terminate the program!

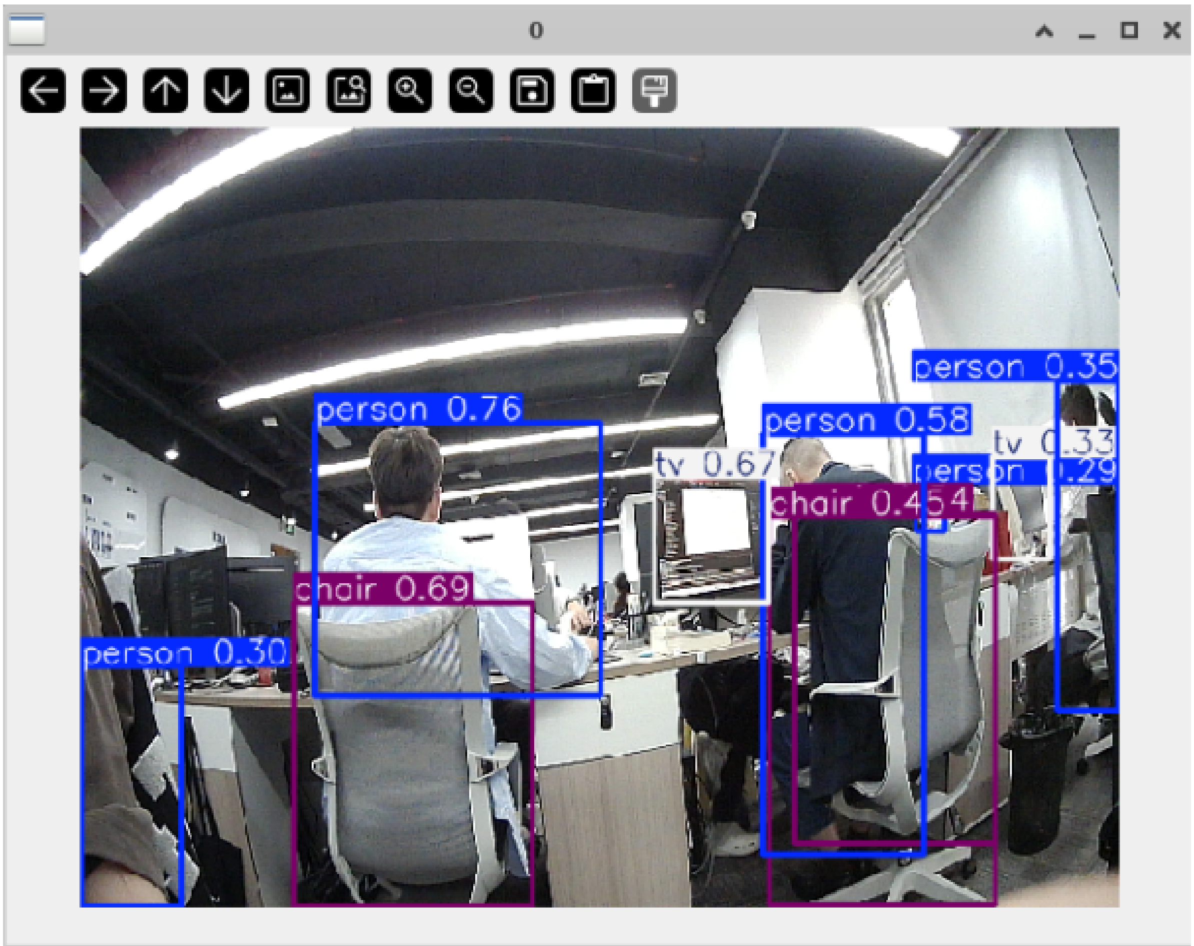
> **Parameter Description**

`model`: Specify YOLO model

`source`: Specify recognition source: multiple cameras can switch numbers

`save=False`: Disable saving results

`show`: Real-time display



References

[Home - Ultralytics YOLO Documentation]

[Command Line Interface - Ultralytics YOLO Documentation]